Weighted cross-scaling and branch composition

Producer proof packet; independent review pending

30 October 2026

Component checked

The explicit UInt64 straight-line cross-scaling computation now connects to the weighted field expressions and the same-support classifier. The final theorem includes the outer equality tests on the two support coefficients: it returns the same optional prelude branch in the word and field descriptions. None means the distinct-support path is entered, not an arithmetic failure.

Literal expressions

For each side use the other point's weights. Compute

$$z_2 = zz \bmod p, \quad z_3 = z_2 z \bmod p, \quad z_4 = z_2 z_2 \bmod p, \quad z_5 = z_4 z \bmod p.$$

The four word outputs are qz_2, rz_4, $(sz_3)w$ and $(tz_5)w$, with remainder after every multiplication. Their decoded values are respectively qz^2, rz^4, sz^3w, tz^5w. All outputs remain canonical. The two sides then compare support coefficients and use inverse-first, equal-second, zero-subtraction-third branch order.

Assumptions and proof mechanism

Every coordinate is canonical modulo p, and the classifier theorem assumes a prime $5 \leq p < 2^{31}$. A finite multiplication-expression induction composes the already checked no-wrap primitive; the final theorem is about the explicit straight-line word function. Nonzero weights and Mumford validity are unnecessary for these polynomial value identities, but remain required by downstream geometric and error-exclusion statements.

Native evidence

Three successive source stages compiled with native exit zero. A separate process checks the final types and four named axiom readbacks. Existing compiler/dependency pins are reused; no new installation, network fetch, scientific trial or cost comparison is performed. Earlier stage sources and receipts remain preserved.

Source correspondence and limits

The source is add22_weighted at projective_law_gate.rs:903–933. Multiplication and square providers are the table-free, n=1 UInt64 field route. The polynomial expression tree is a proof device; no operation-count claim is made from its evaluator or from Lean evaluation order. Counter side effects, checked-panic semantics, compiler correctness, reachability, and delegated branch payloads remain separate. In particular sameUDouble identifies delegation to double22_weighted, not its eventual returned label. The final theorem establishes value composition and admission to the same-support branch, not complete addition-law correctness.